PM - IMP PYQ Analysis

PROJECT MANAGEMENT | BE05000481 | Sem 5 | GTU
New Syllabus 2024-25 | Papers Analyzed: 9 (W21-25)

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Chapter Reference (OLD → NEW Mapping)

New Ch.	Covers	Status
Ch. 1	Project Management: definition, goals, life cycle, portfolio process, formulation	✅ Valid
Ch. 2	Organizing + Planning: PM roles, org structures, WBS, project plan	✅ Valid
Ch. 3	Budgeting: methods, cost estimating, budget uncertainty & risk	✅ Valid
Ch. 4	Project Network Theory: PERT/CPM, Gantt, resource loading/leveling, crashing *(merged old Ch.5 + Ch.6)*	🔶 Modified
Ch. 5	Monitoring & Controlling + Evaluation/Termination *(merged old Ch.7 + Ch.8)*	🔶 Modified

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📌 M.IMP Questions - High → Low Priority

> **Priority Rule:** Freq ≥ 4 OR (Freq ≥ 3 AND Max Marks = 7) → M.IMP

1. **CPM/AON Network Construction - Critical Path (ES, EF, LS, LF)**
   `(Ch. 4)` | **Freq: 6x** | **Max Marks: 7** | `Num: 6`
2. **Gantt Chart (Explain / Construct)**
   `(Ch. 4)` | **Freq: 6x** | **Max Marks: 7** | `Theory: 5` `Diag: 1`

3. **Budget Uncertainty & Risk Management (Risk Identification/Analysis)**
   `(Ch. 3)` | **Freq: 6x** | **Max Marks: 7** | `Theory: 6`
4. **Contents of a Project Plan & Planning Process Overview**
   `(Ch. 2)` | **Freq: 6x** | **Max Marks: 7** | `Theory: 6`
5. **Earned Value Analysis (EVA) / Earned Value Chart**
   `(Ch. 5)` | **Freq: 5x** | **Max Marks: 7** | `Theory: 4` `Num: 1`
6. **Resource Loading (Explain / Factors Affecting)**
   `(Ch. 4)` | **Freq: 5x** | **Max Marks: 7** | `Theory: 5`
7. **Budgeting Methods (Top-down / Bottom-up)**
   `(Ch. 3)` | **Freq: 5x** | **Max Marks: 7** | `Theory: 5`
8. **Work Breakdown Structure / WBS (Gozinto Tree)**
   `(Ch. 2)` | **Freq: 5x** | **Max Marks: 7** | `Theory: 3` `Diag: 2`
9. **Project Management vs. General Management**
   `(Ch. 1)` | **Freq: 5x** | **Max Marks: 7** | `Theory: 5`
10. **Project Portfolio Process (Steps Involved)**
    `(Ch. 1)` | **Freq: 4x** | **Max Marks: 7** | `Theory: 4`
11. **Roles & Responsibilities of Project Manager**
    `(Ch. 2)` | **Freq: 4x** | **Max Marks: 7** | `Theory: 4`
12. **Mechanisms / Types of Project Control & System Design**
    `(Ch. 5)` | **Freq: 4x** | **Max Marks: 7** | `Theory: 4`
13. **Scope Creep (Meaning + Causes)**
    `(Ch. 5)` | **Freq: 4x** | **Max Marks: 4** | `Theory: 4`
14. **Three Primary Goals of a Project (Cost, Time, Scope/Quality)**
    `(Ch. 1)` | **Freq: 4x** | **Max Marks: 4** | `Theory: 4`
15. **Project Team (Roles, Characteristics & Multidisciplinary Teams)**
    `(Ch. 2)` | **Freq: 4x** | **Max Marks: 4** | `Theory: 4`
16. **Project Crashing (Cost/Time Trade-off Numerical)**
    `(Ch. 4)` | **Freq: 3x** | **Max Marks: 7** | `Num: 3`
17. **Fulkerson's Rule for Node Numbering**
    `(Ch. 4)` | **Freq: 2x** | **Max Marks: 7** | `Theory: 1` `Num: 1`

IMP Questions - High → Low Priority

1. **Weighted Scoring Model (Project/Location Selection Numerical)**
   `(Ch. 1)` | Freq: 3x | Max Marks: 7 | `Num: 3`
2. **Project Organization Types (Matrix / Functional / Pure)**
   `(Ch. 2)` | Freq: 3x | Max Marks: 7 | `Theory: 3`
3. **Project Audit - Steps / Objectives / Report Contents**
   `(Ch. 5)` | Freq: 5x | Max Marks: 4 | `Theory: 5`
4. **Project Termination - Types / Process**
   `(Ch. 5)` | Freq: 4x | Max Marks: 7 | `Theory: 4`
5. **Goldratt's Critical Chain**
   `(Ch. 4)` | Freq: 3x | Max Marks: 7 | `Theory: 3`
6. **Data Collection Methods / Advantages of Data Reporting**
   `(Ch. 5)` | Freq: 4x | Max Marks: 4 | `Theory: 4`
7. **Cost Estimating / Improving Cost Estimates**
   `(Ch. 3)` | Freq: 4x | Max Marks: 4 | `Theory: 4`
8. **FMEA - Short Note / RPN / Severity-Likelihood-Detectability**
   `(Ch. 3)` | Freq: 3x | Max Marks: 4 | `Theory: 2` `Num: 1`
9. **Life Cycle Types (S-type vs J-type) & Resource Borrowing**
   `(Ch. 1)` | Freq: 3x | Max Marks: 4 | `Theory: 3`
10. **Plan-Monitor-Control Cycle**
    `(Ch. 5)` | Freq: 3x | Max Marks: 4 | `Theory: 3`
11. **Learning Curve Costing Numerical (Bid Price / Budget Estimation)**
    `(Ch. 3)` | Freq: 2x | Max Marks: 7 | `Num: 2`
12. **RACI Matrix**
    `(Ch. 2)` | Freq: 2x | Max Marks: 7 | `Theory: 2`
13. **Simulation (Monte Carlo / Steps of Simulation Process)**
    `(Ch. 4)` | Freq: 2x | Max Marks: 7 | `Theory: 1` `Def: 1`
14. **PERT/CPM Definition & Application (Conceptual)**
    `(Ch. 4)` | Freq: 2x | Max Marks: 7 | `Theory: 2`

15. **Project Buffer / Feeding Buffer / Resource Buffer (Define)**
    `(Ch. 4)` | Freq: 2x | Max Marks: 3 | `Def: 2`
16. **Selection Criteria for a Project Manager**
    `(Ch. 2)` | Freq: 2x | Max Marks: 3 | `Theory: 2`
17. **Resource Leveling (Explain)**
    `(Ch. 4)` | Freq: 3x | Max Marks: 4 | `Theory: 3`
18. **PERT Probability / Three-Time-Estimate Numerical**
    `(Ch. 4)` | Freq: 1x | Max Marks: 4 | `Num: 1`
19. **AON and AOA Representation (Define / Differentiate)**
    `(Ch. 4)` | Freq: 1x | Max Marks: 3 | `Theory: 1`

⚫ New Syllabus Blind Spots - High → Low Risk

Risk	Topic	Chapter
🔴 **HIGH**	Abandonment Analysis	Ch. 5
🔴 **HIGH**	Time Scale Version of Network	Ch. 4
🔴 **HIGH**	Milestone Chart vs. Gantt Chart	Ch. 4
🟠 **MED-HIGH**	Multi-Project Scheduling and Resource Allocation	Ch. 4
🟠 **MED-HIGH**	Extensions to PERT/CPM	Ch. 4
🟡 **MEDIUM**	Constrained Resource Scheduling	Ch. 4
🟡 **MEDIUM**	Confronting Uncertainty (general)	Ch. 1
🟢 **LOW-MED**	WBS + Responsibility Matrix Integration	Ch. 2

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